

Hopf-Link Solutions and the Rope Soliton Spectrum

Stable Knotted Solitons of the Rope Field, Computed and Reproduced

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Abstract

The rope field supports stable, localised knotted solutions — solitons whose stability is topological. This paper reports the computed spectrum of the simplest such solutions. A single ring knot relaxes to a stable equilibrium at radius $R^* = 0.856$ (in the natural units of the solver) with minimum energy $E_{\min} = 12.08 M_{\text{Pl}}$; a Hopf link relaxes to $R^* = 0.838$. These are reproduced results, stable across the relaxation solver and its regression tests. All numbers in this paper are outputs of the `rope_solver` computation in this package (reproducible via `benchmarks/reproduce_results.py`), not inserted constants.

Scope of this paper

CLAIM: the rope field admits stable knotted solitons with a computed spectrum (ring $R^*=0.856$, $E=12.08 M_{\text{Pl}}$; Hopf link $R^*=0.838$), whose stability follows from a conserved linking number.

NOT CLAIMED: that these solitons ARE the observed particles, nor a derivation of particle masses from them (the mass-ratio question is treated separately and honestly, with its failures pinned). This paper reports the soliton spectrum as computed classical field solutions.

1. Knotted Solutions of the Rope Field

A rope configuration carrying a non-trivial topology cannot relax to the trivial vacuum without cutting a strand, which the dynamics forbid. The configuration therefore settles into a localised minimum-energy knot — a soliton whose existence is guaranteed by topology rather than by a delicate balance of forces. The simplest cases are the single ring and the Hopf link.

2. The Computed Spectrum

Relaxing each configuration in the rope solver yields a definite equilibrium radius and energy:

Solution	Equilibrium R^*	Min. energy	Linking
Single ring knot	0.856	12.08 M_{Pl}	self-linked
Hopf link (sourced)	0.838	—	$Lk = 1$

Both values are stable reproductions (published $R^* = 0.85$ and 0.83 respectively; the solver returns 0.856 and 0.838 within tolerance). The ring energy $12.08 M_{\text{Pl}}$ is likewise a reproduced figure. These are the ground-state entries of the rope soliton spectrum.

Status

Stable ring knot $R^*=0.856$, $E=12.08 M_{\text{Pl}}$ — Modeled (reproduced numerically).

Hopf link $R^*=0.838$ — Modeled (reproduced numerically).

Identification with physical particles — Open, not claimed here.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.